Readout system based on 128 channels FPGA-TDC system for FPMT TOF-PET

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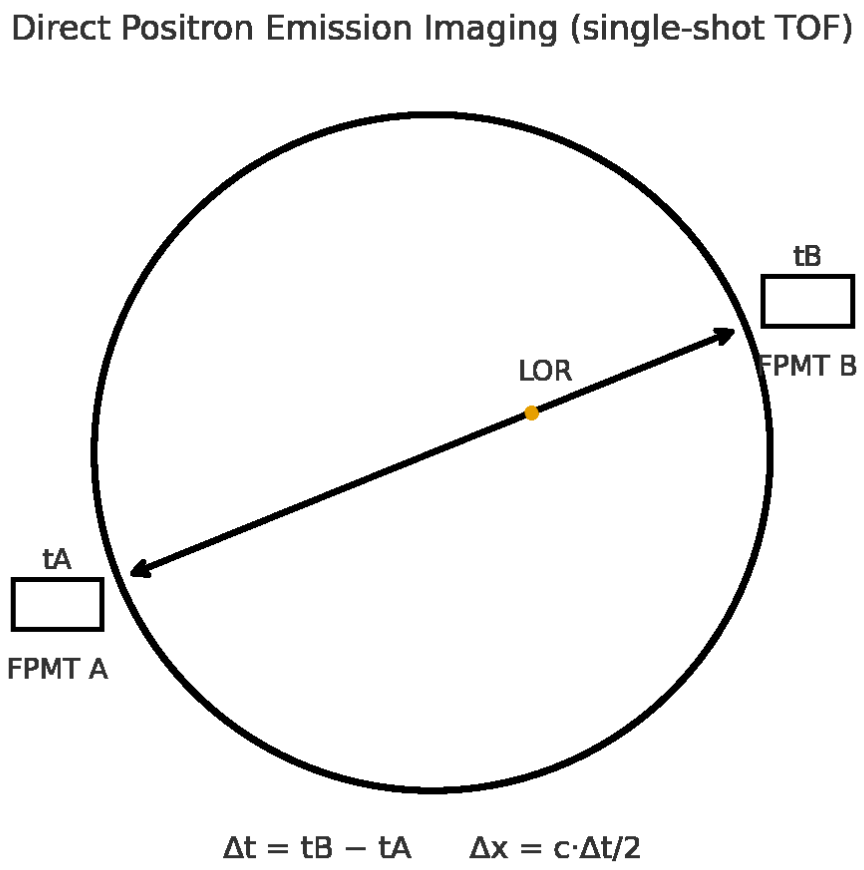
On behalf of electronics group, FPMT collaboration group

Overview

1. Introduction
2. FPGA TDC hardware
3. FPGA TDC firmware
4. Measurements
5. FPMT imaging

Introduction

- Cherenkov photons can provide more precise timestamps;
- Single photoelectrons will be detected when using Cherenkov radiator window;
- MCP-PMT is good at detecting SPE when requiring a high-resolution timing

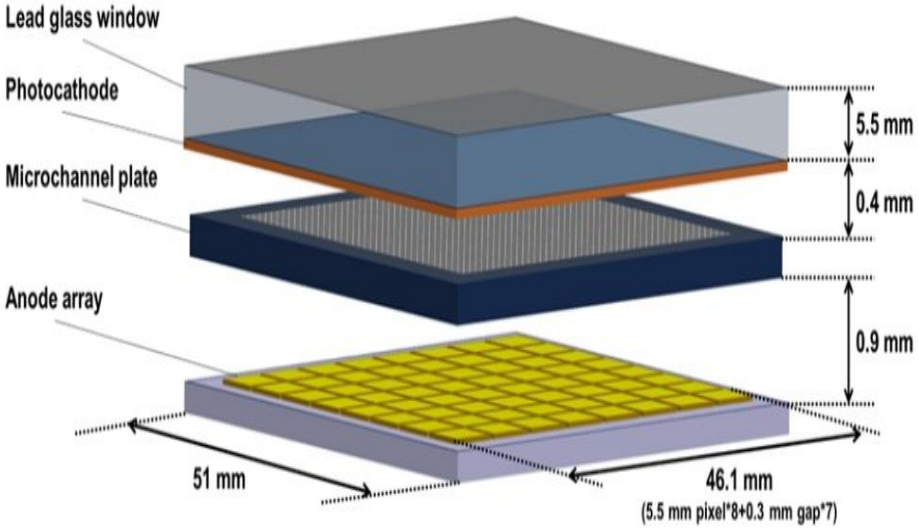


direct positron emission imaging
This picture is generated by AI

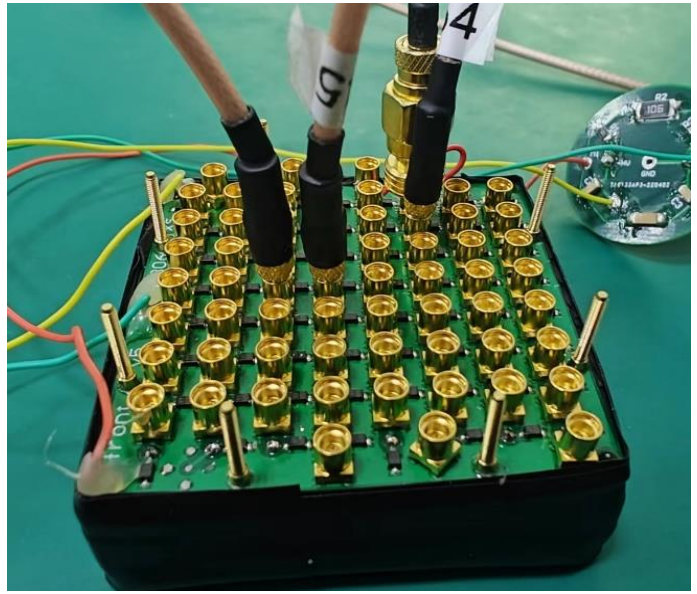
NNVT FPMT Introduction

- Gain: 4E+6
- Amplitude: 50 mV
- Rising time: 300 ps
- Pulse width: 1 ns
- More details, please see ref[1] and ref[2]

FPMT front side



FPMT structure



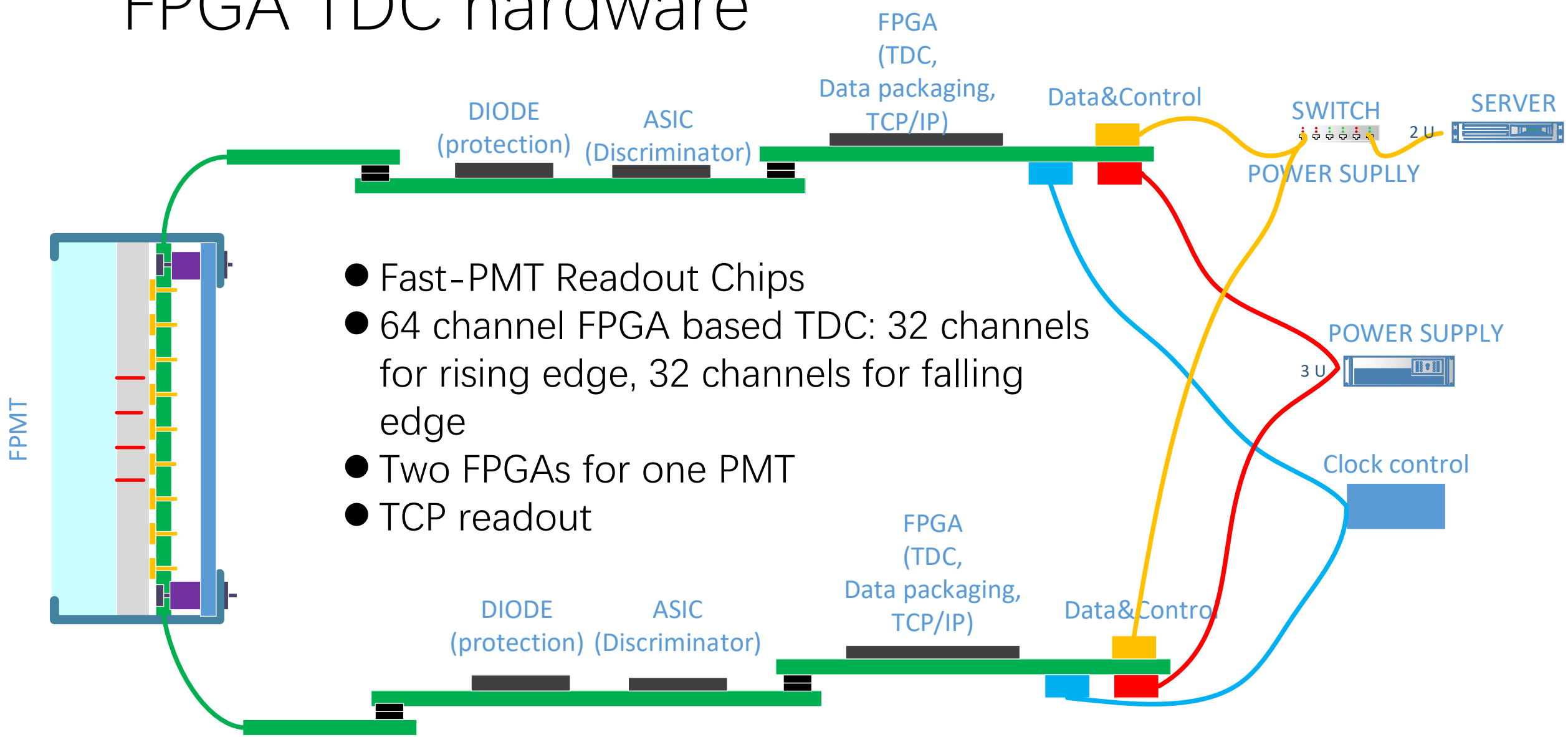
FPMT back side with a readout board

Introduction

Electronics Requirements:

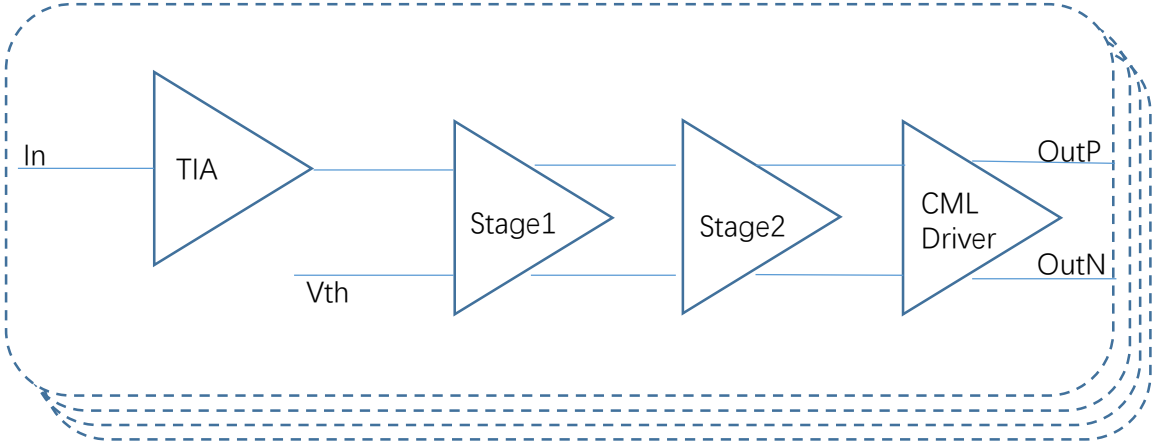
- 64 readout channels for one FPMT
- Amplify the signal
- Independent threshold adjustment
- ~10 ps CTR (Coincidence Time Resolution)
- Support dense packing of FPMT modules

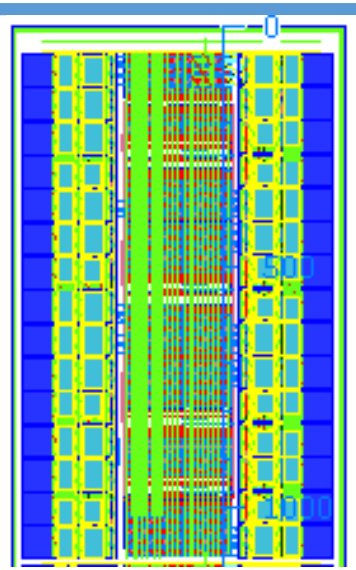
FPGA TDC hardware

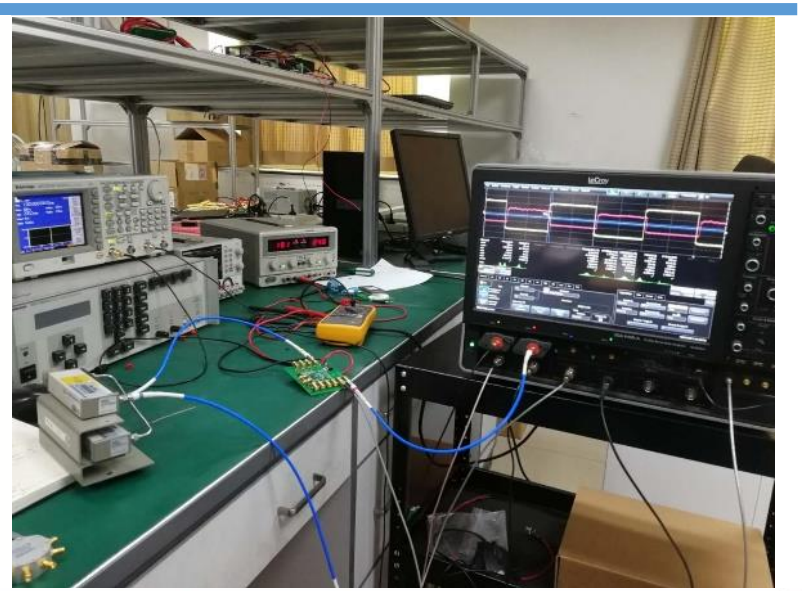


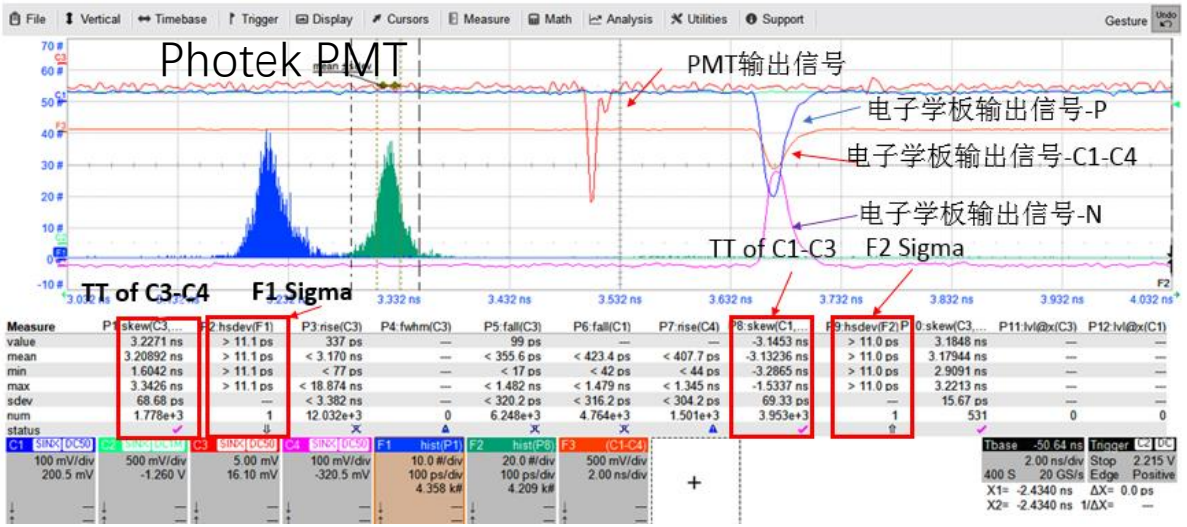
- Fast-PMT Readout Chips
- 64 channel FPGA based TDC: 32 channels for rising edge, 32 channels for falling edge
- Two FPGAs for one PMT
- TCP readout

Fast-PMT Readout Chip



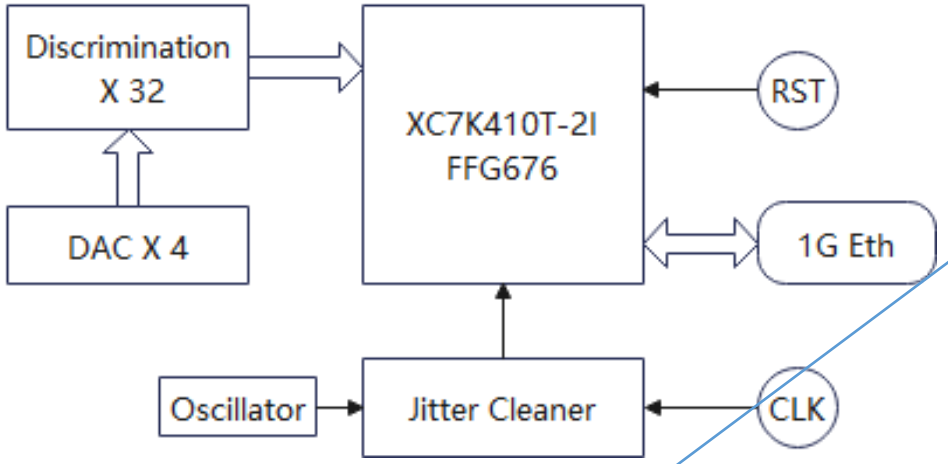




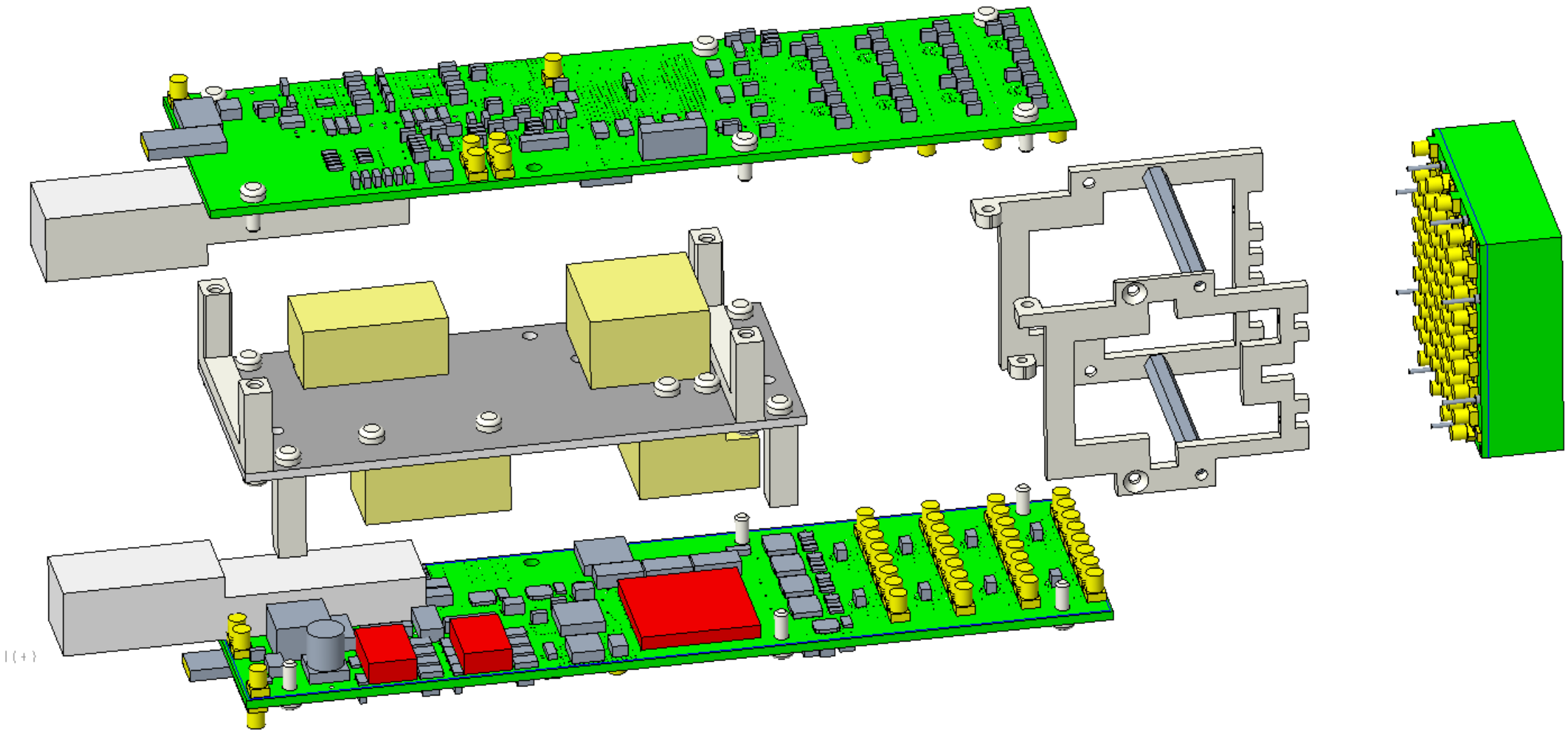


- 4 channels chip for amplication and discrimination
  - SMIC 130nm
  - TIA with high bandwidth
  - CML driver
- The chip is wire bonded to PCB
- Output Jitter < 5ps with Photek PMT210 @ SPE
- 6 times amplifications

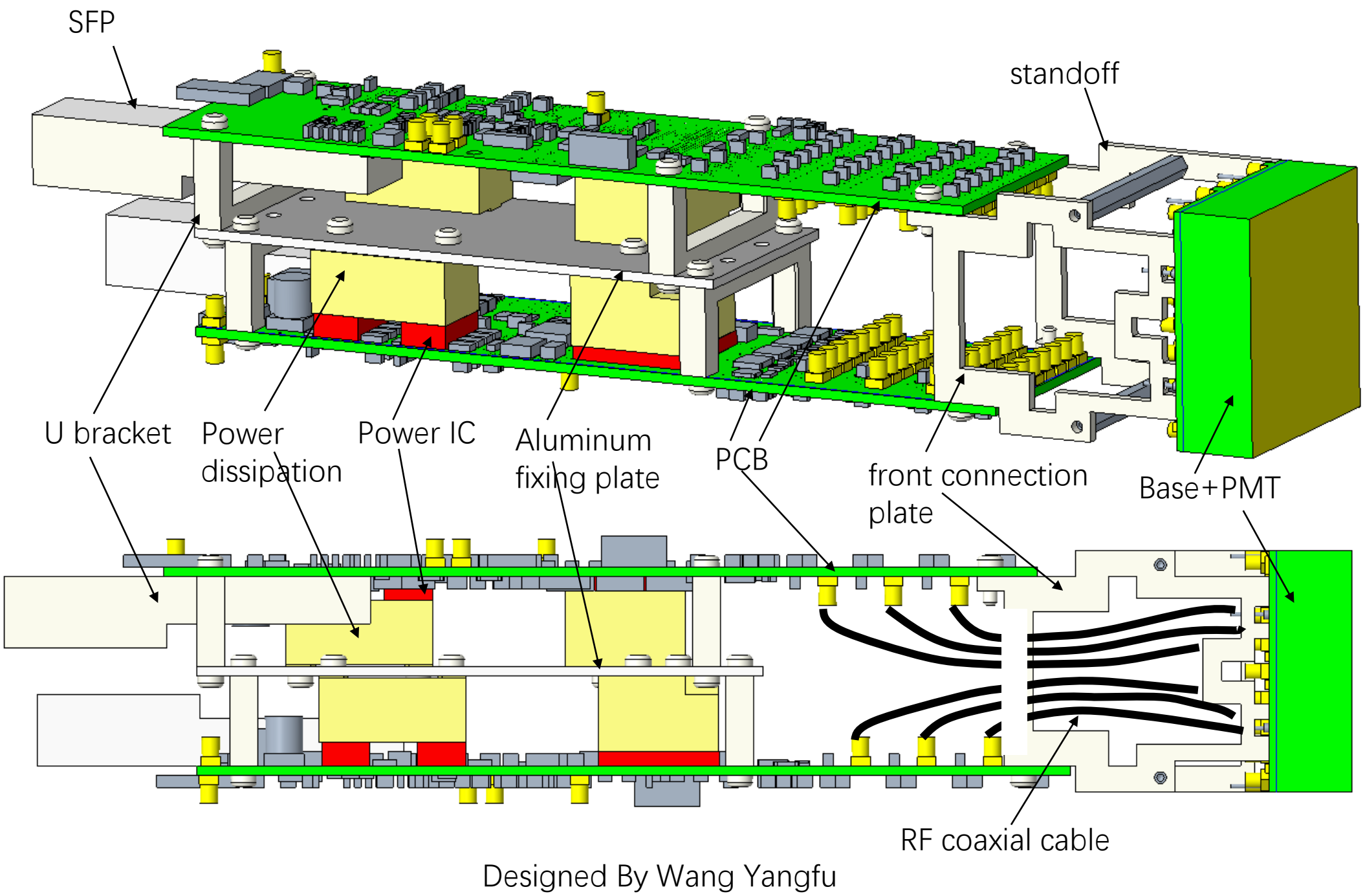
TDC boards



- FPGA-TDC boards, Designed by Hu Jun
- 8 Fast-PMT Readout chips on boards, Designed by Yan Xiongbo
- 32 channels for PMT input
- 64 TDC channels for ToA and ToT;
- TCP readout

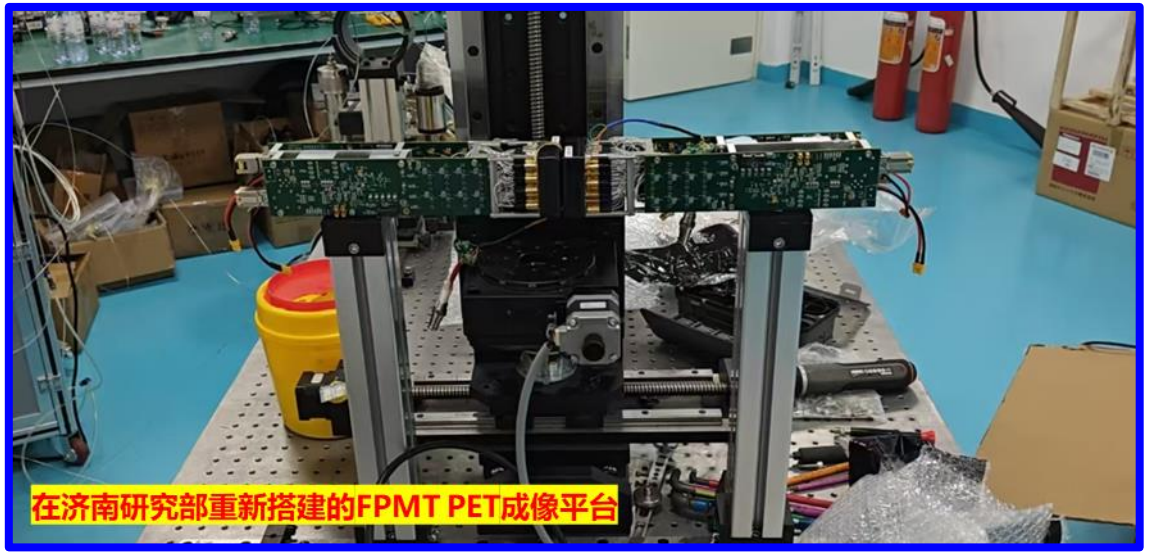


Designed By Wang Yangfu

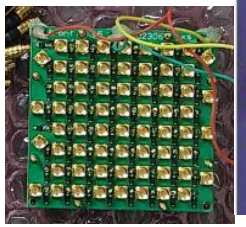


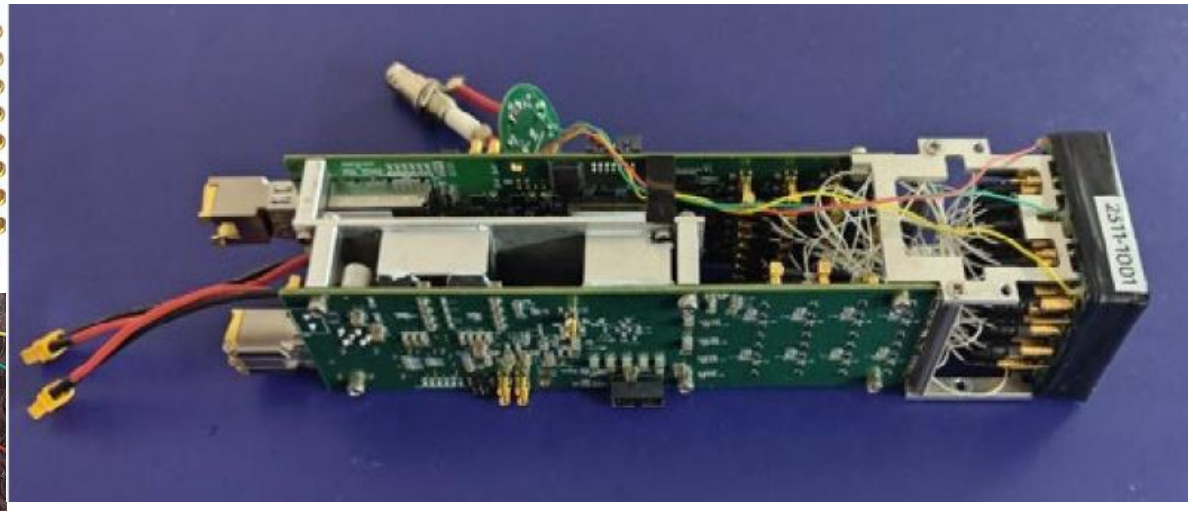
Designed By Wang Yangfu

FPMT PET Imagine System

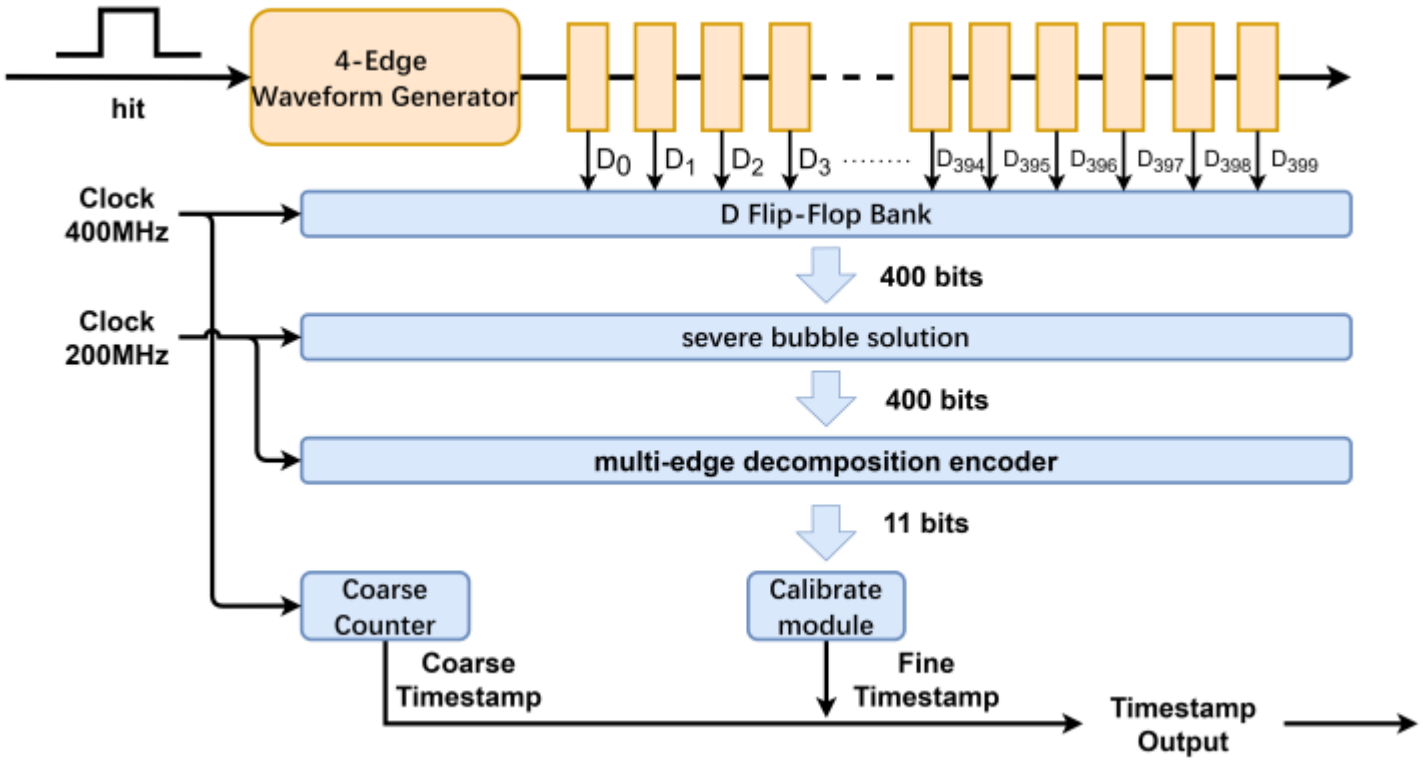


- Pb-Glass MCP-PMT
- TDC board: timing
- Rst/Calibration board: Counter Synchronization
- Power Supply System
- Si5361*1: clock distirbution





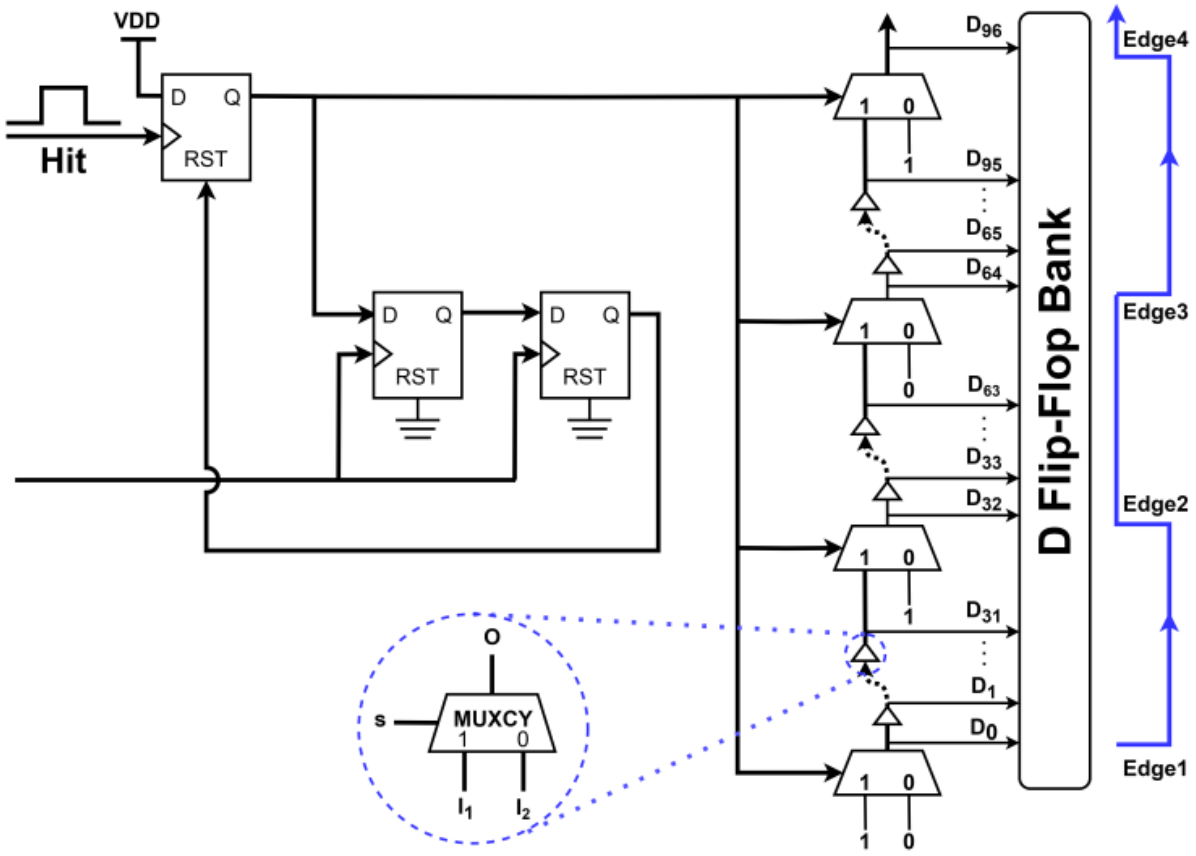
FPGA TDC firmware



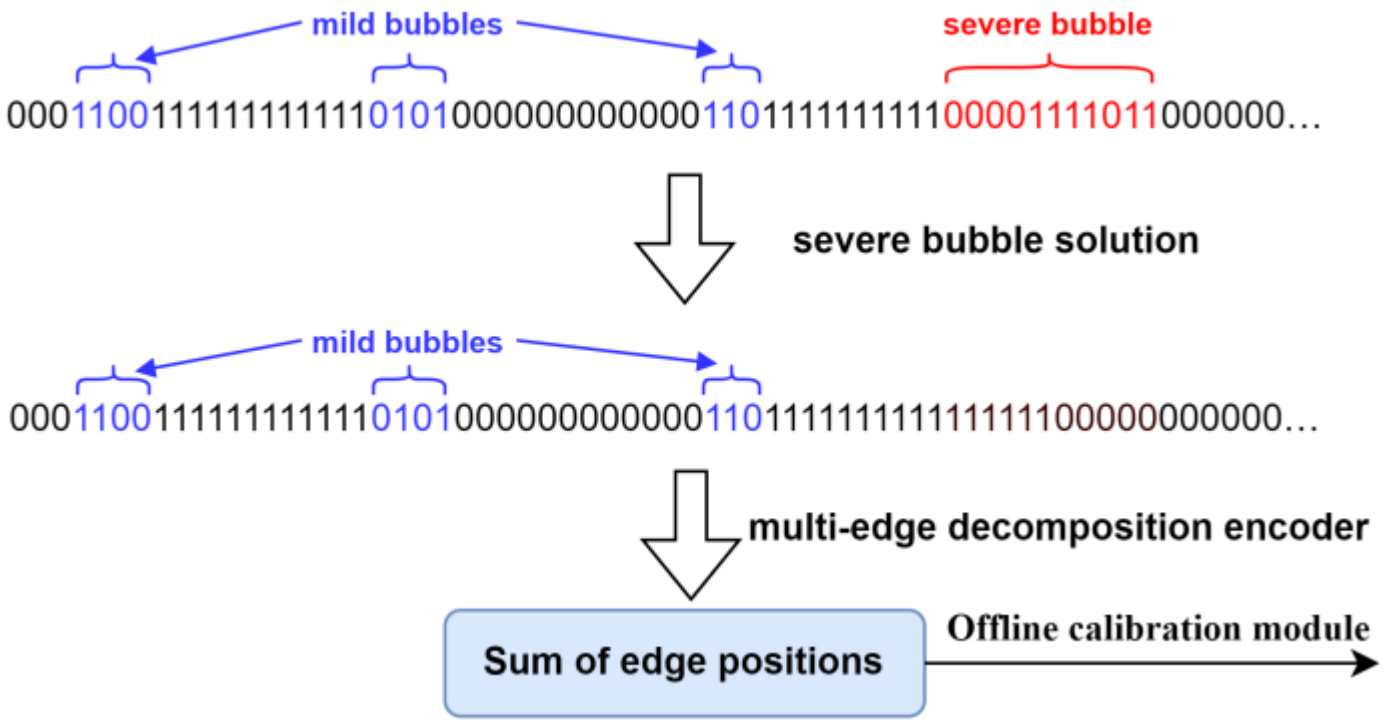
Structure of the 4-edge TDL TDC: ref[3]

Structure of the Wave Generator

- One TDL has 100 CARRY4 with 400 taps
- Edge position: 0, 32, 64, and 96
- 3 clock cycles of dead time

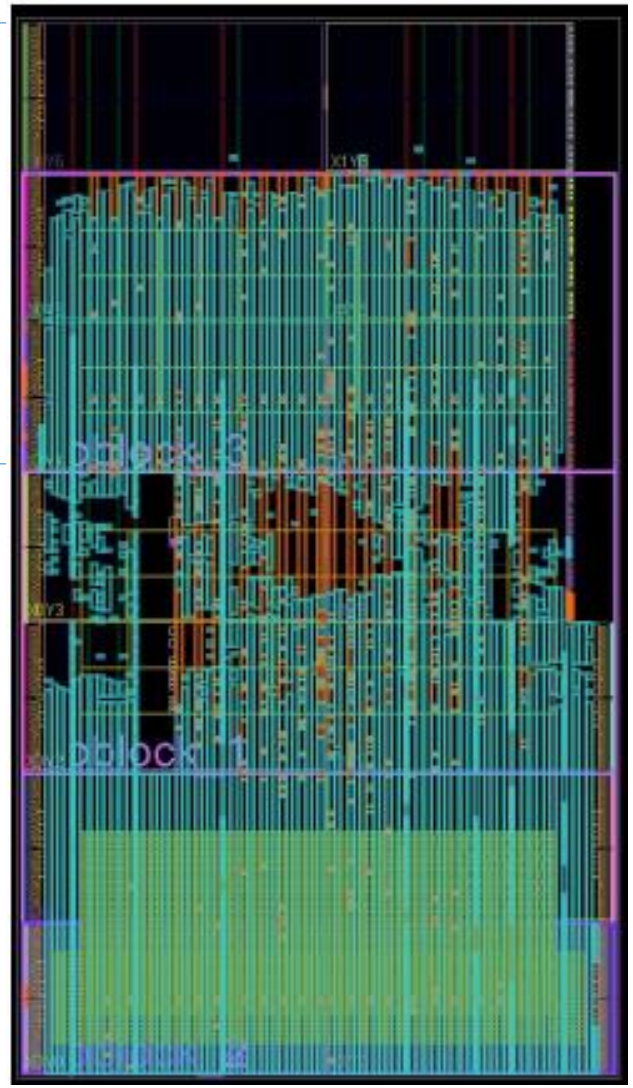


overall flow of de-bubble process



Severe bubble zone

mild bubble zone

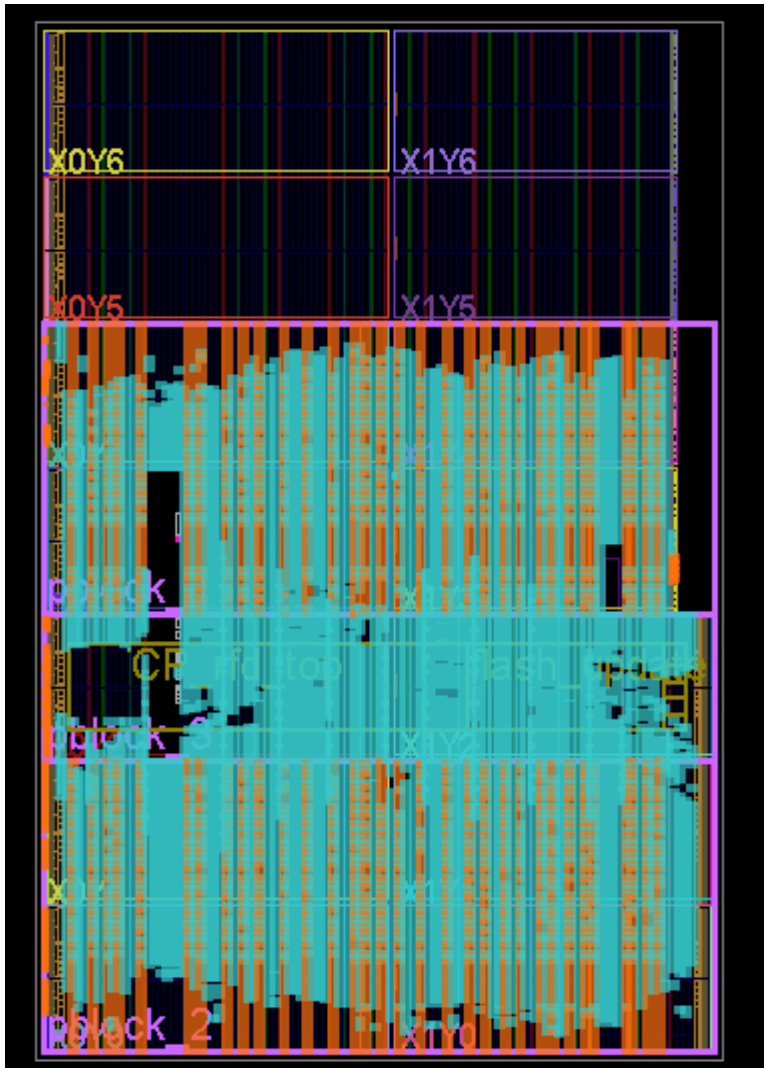


Optimize layout

- The test shows that the zone from XmY5 and above will introduce severe bubbles, and try to put all the TDLs in the mild bubbles zone which will save much resource for de-bubble;
- X0Y4 and X1Y4 are not severe zones after checking

Severe bubble zone

mild bubble zone



Used Resources

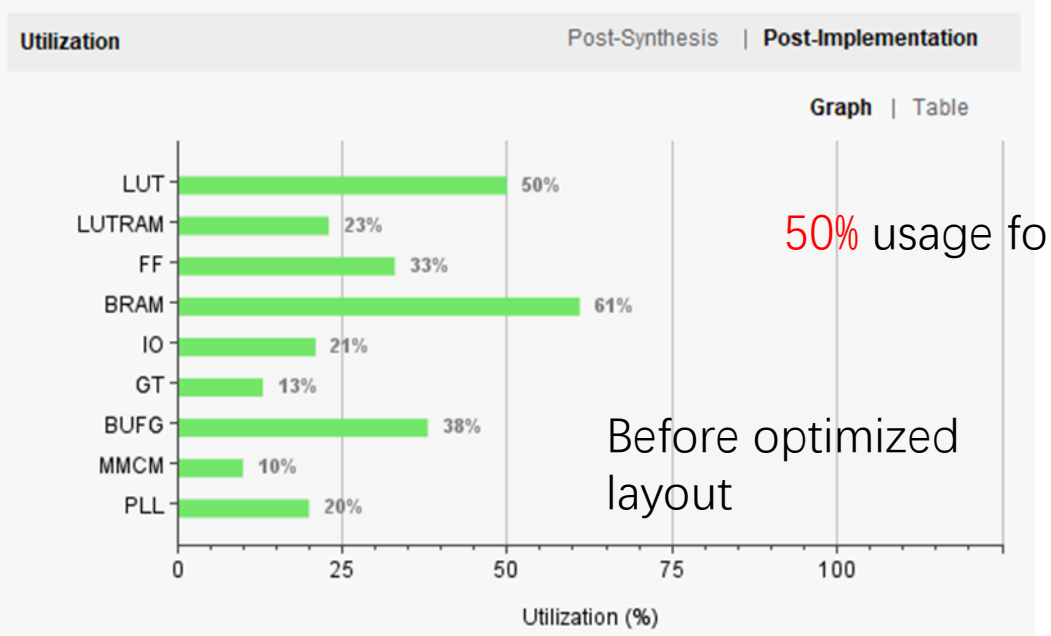
- 32 channels for rising edge, 32 channels for falling edge
- Kintex-7 410t
- Clock: 400 MHz
- LUT: 38%

Timing

Worst Negative Slack (WNS):	0.066 ns
Total Negative Slack (TNS):	0 ns
Number of Failing Endpoints:	0
Total Number of Endpoints:	170039

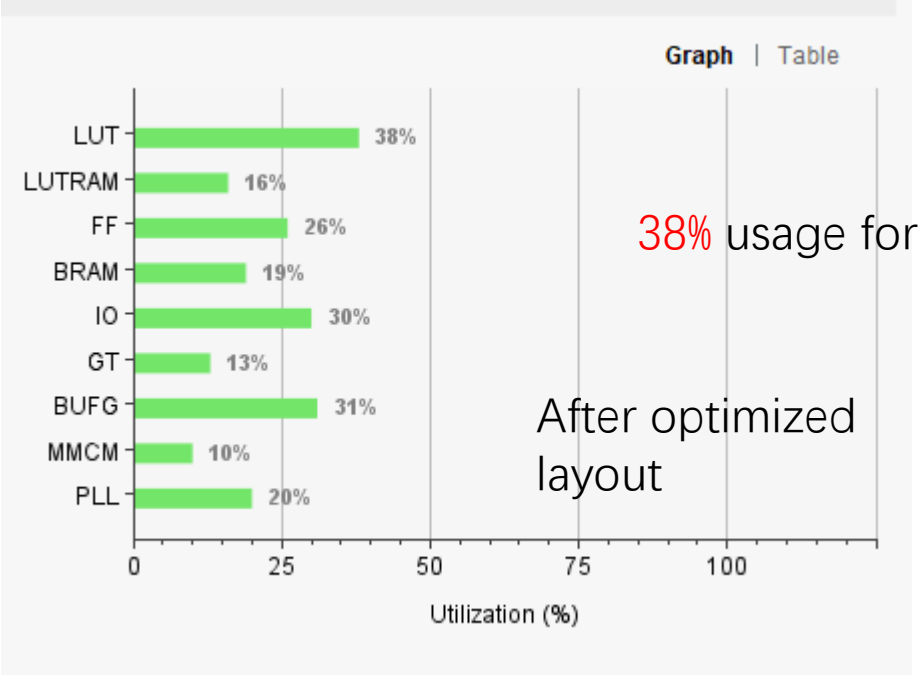
Implemented Timing Report

Timing closure



50% usage for LUT

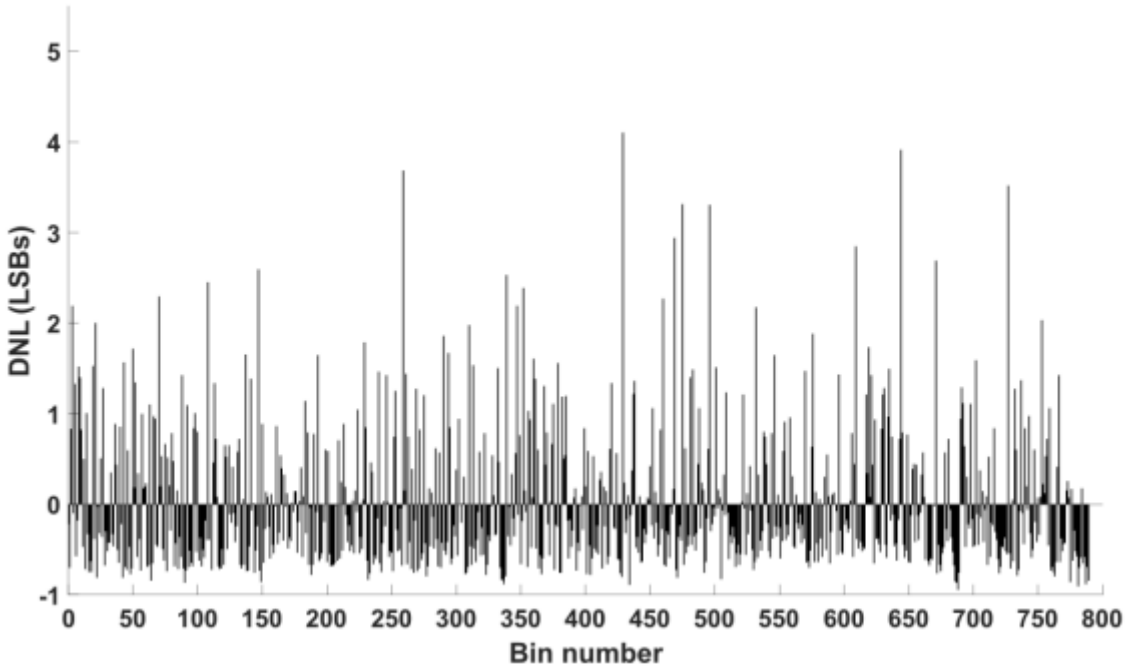
Before optimized layout

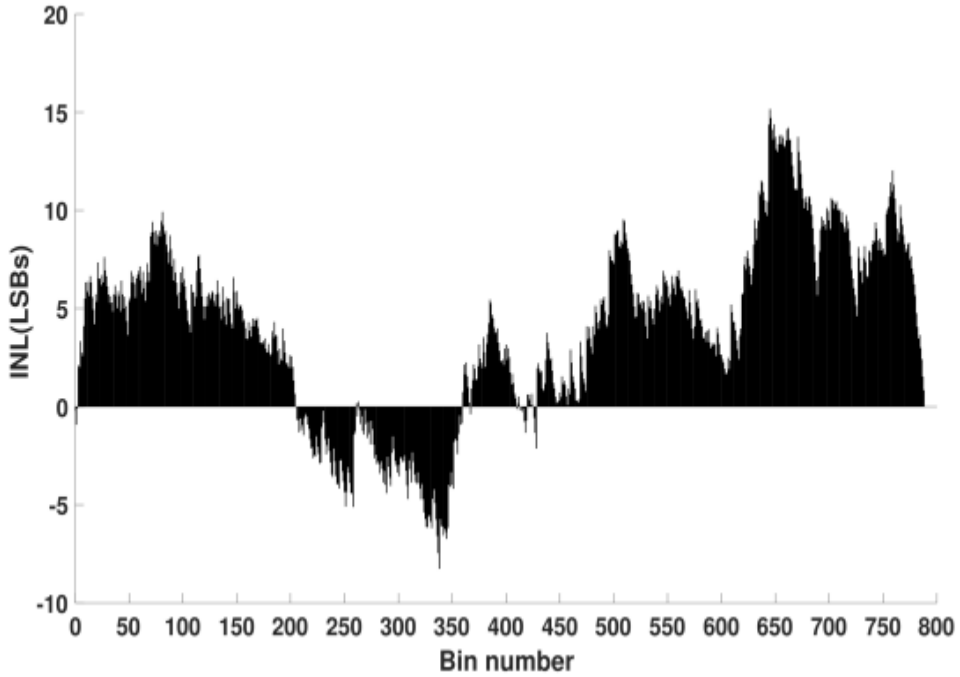


38% usage for LUT

After optimized layout

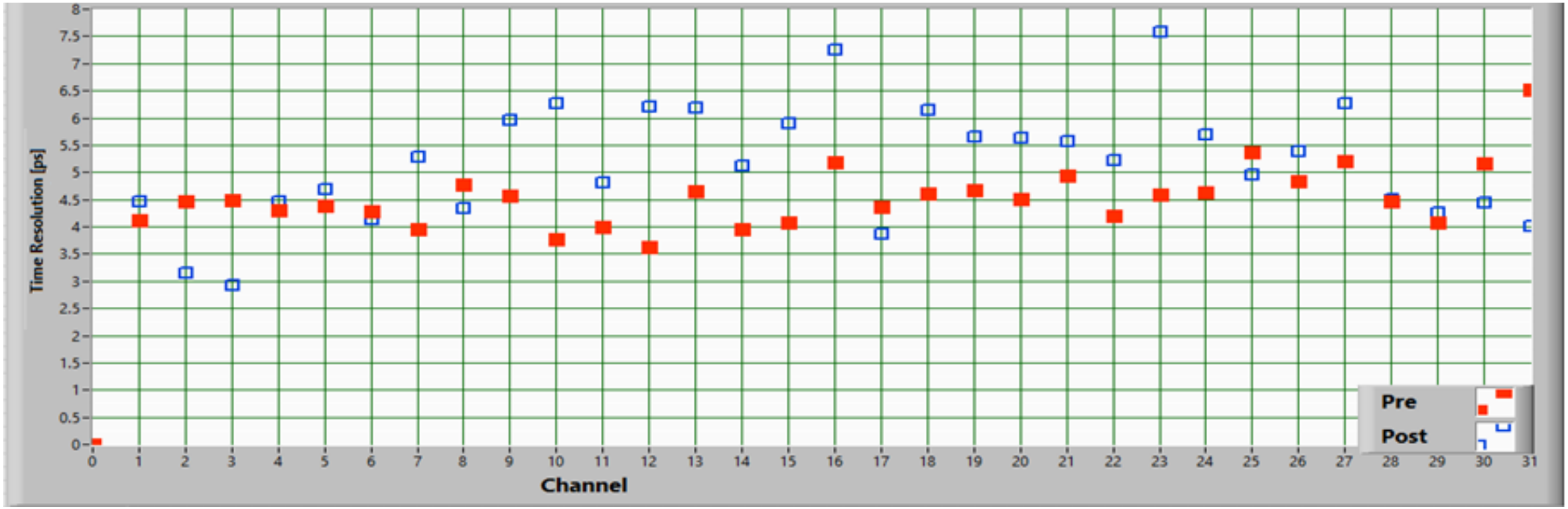
DNL & INL





FPGA temperature was 46.6°C

CTR in a board



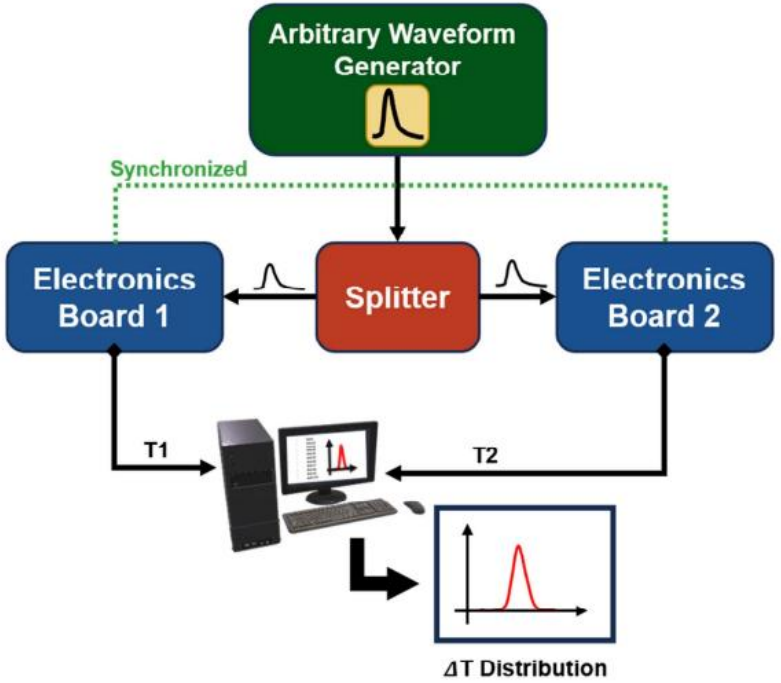
CTR = ~4.5 ps, only test for TDC

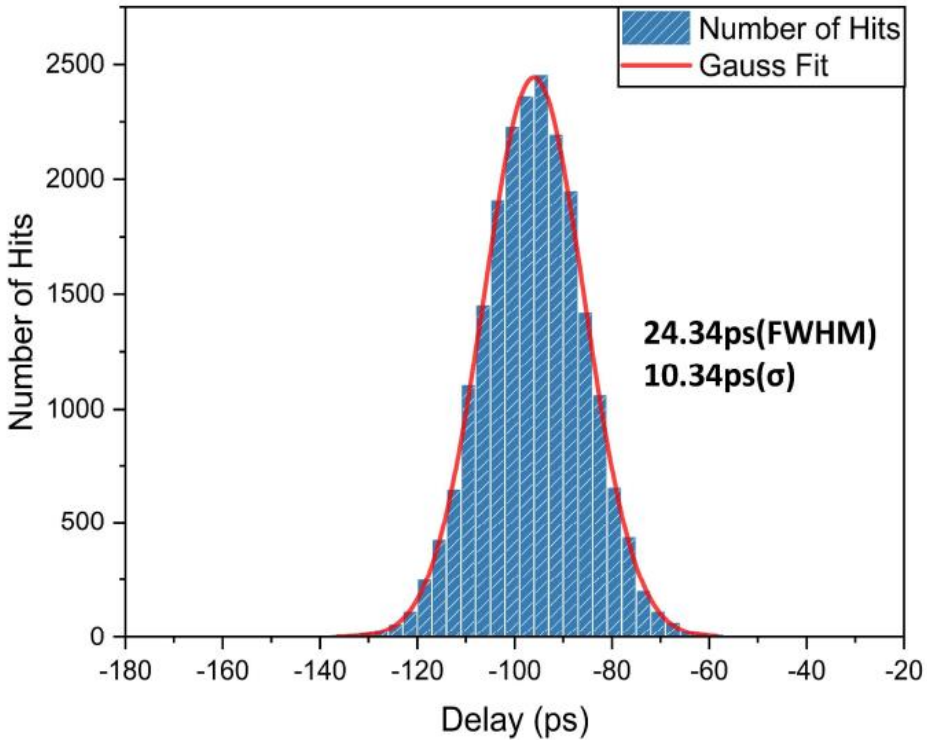
Measurements based on a generator

Keysight M8195A, 65 Gsa/s

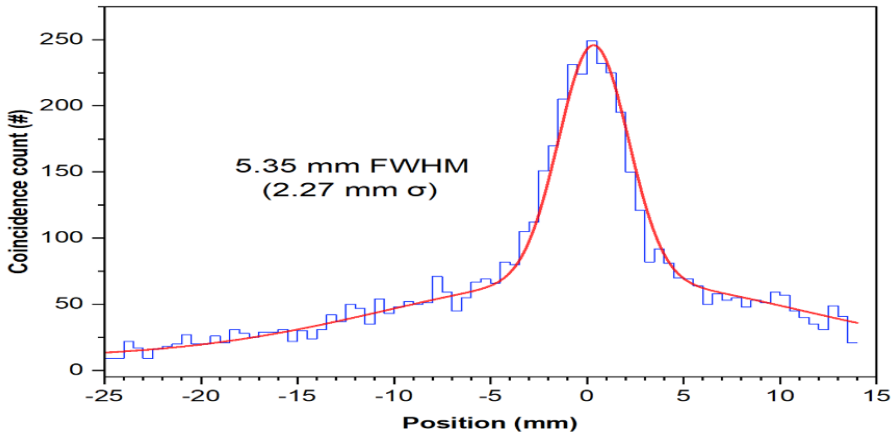
500 ps falling time， 300 ps Rise time for simulating FPMT signal

Inject it into Fast-PMT Readout chips

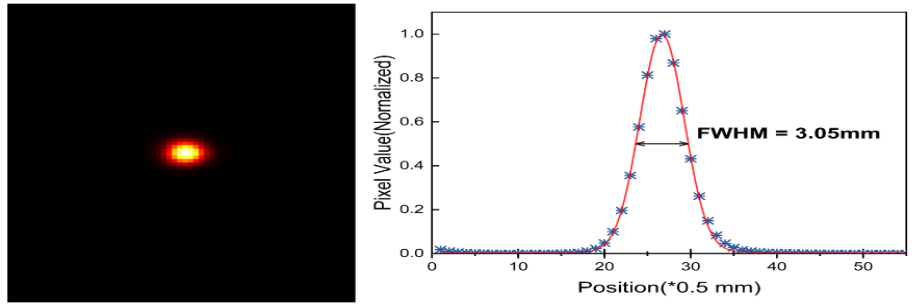




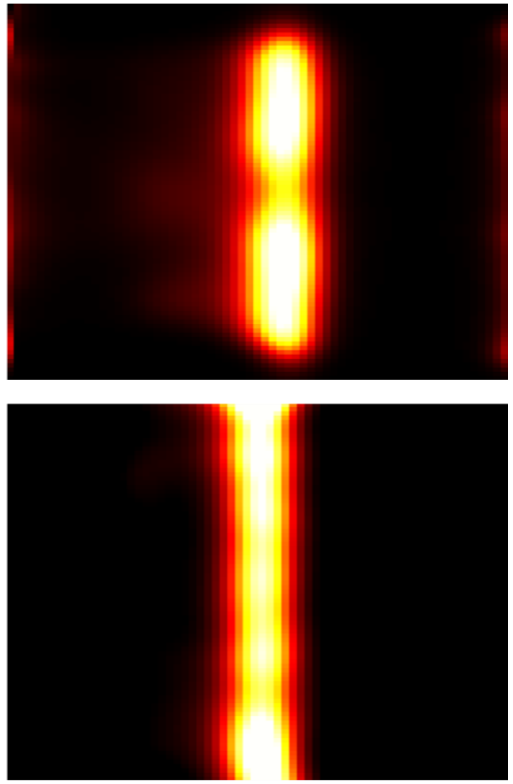
point source & rod source imaging



(a)



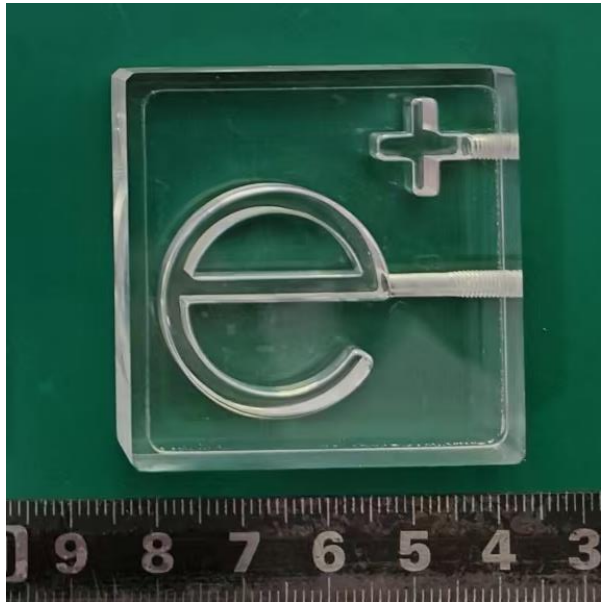
(b)

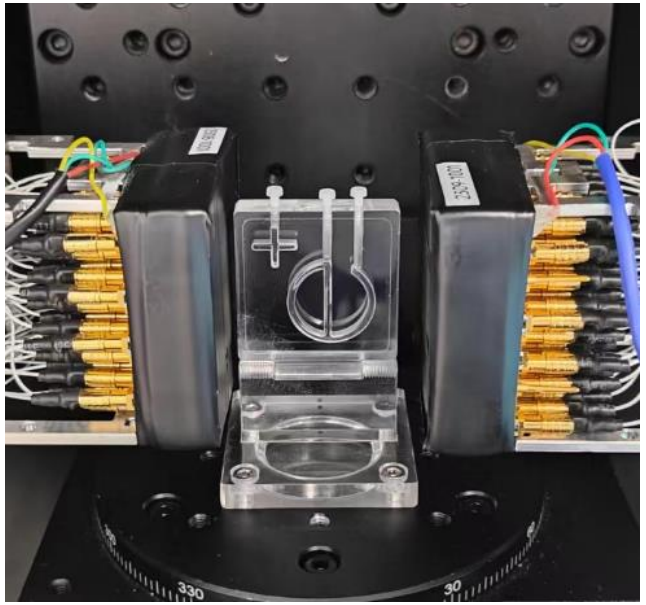


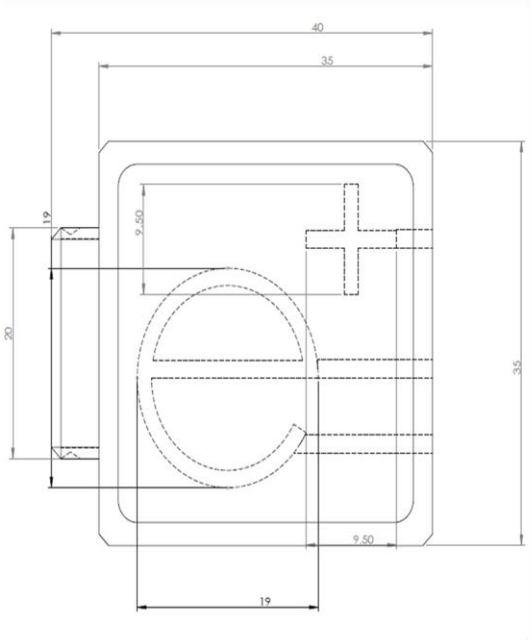
(c)

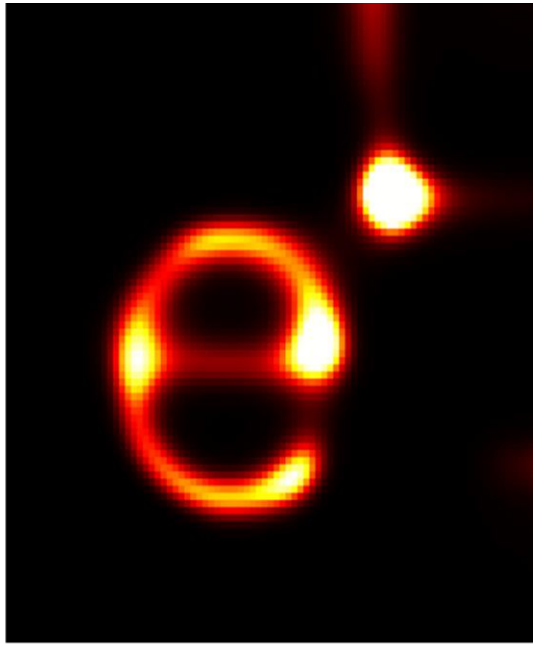
- Result of the intrinsic resolution test (a) and **point source image** results and analysis (b), rod source image before and after normalization correction (c). [2]

phantom imaging









~200 µCi of ^{18}F ionic solution was injected into the phantom

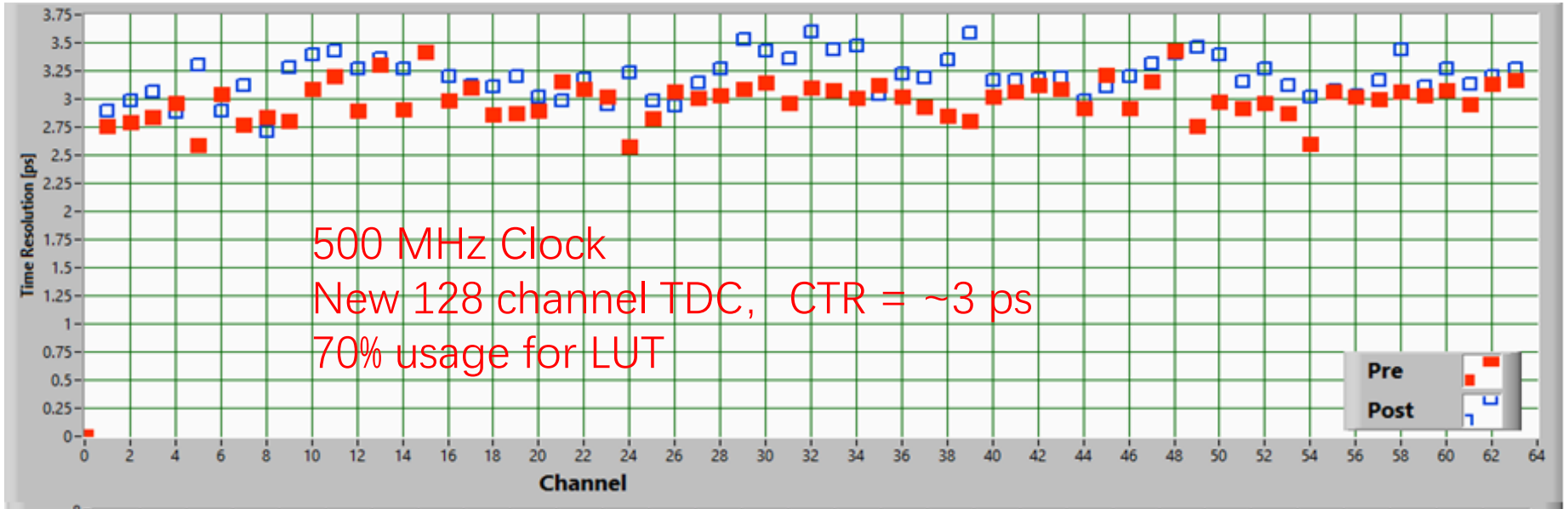
Summary

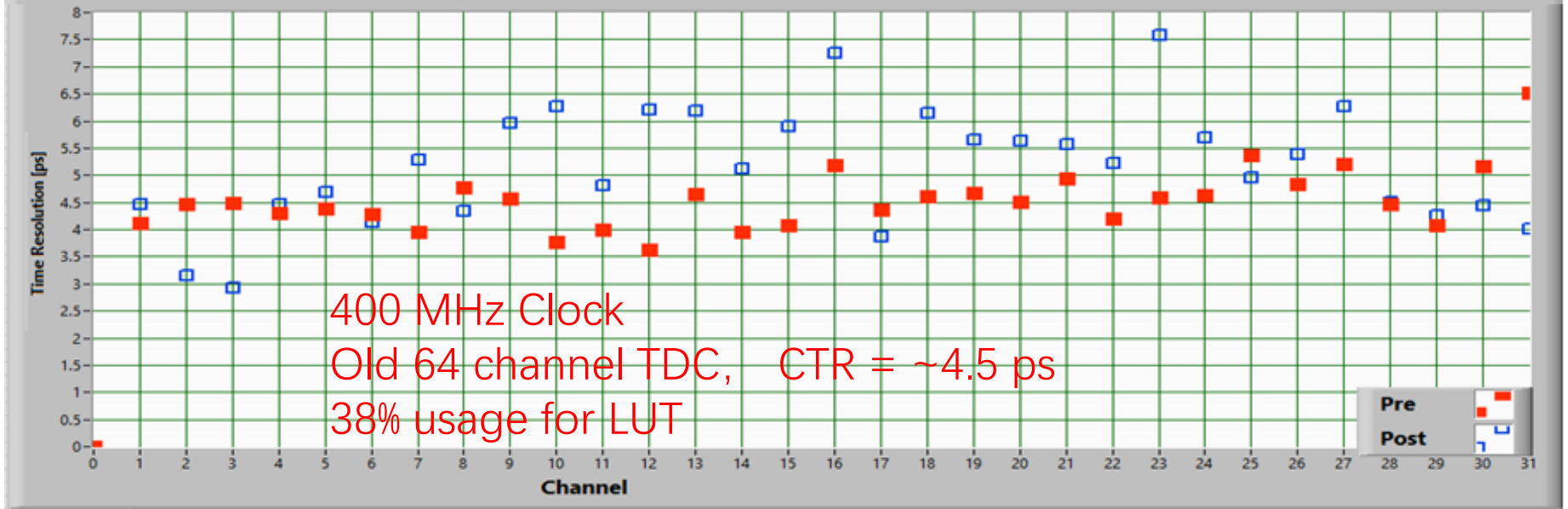
- 32 channels readout boards had been used: two FPGAs, one PMT;
- 6 times amplifications;
- 10 ps CTR goal had been reached;
- supporting Independent threshold adjustment
- supporting dense packing of FPMT modules

Next steps

- More tests should be done such as TDC performance VS temperature variation;
- More TDC channels in FPGA: one FPGA, one FPMT;
- Lower analog noises: more distances between readout chips and FPGA
- More readout boards for large size imaging

New FPGA TDC





Same FPGA

Kintex-410t

Higher CTR

Lower distribution

Ref

[1] Chen L, Ma L, et al 2024 Coincidence time resolution of 50 ps FWHM using a pair of multi-anode MCP-PMTs with Cherenkov radiator window *Nucl. Instruments Methods Phys. Res. Sect. A Accel. Spectrometers, Detect. Assoc. Equip.* 1062

[2] Weiyan Pan et al., arXiv:2502.07195

[3] Zehong Liang et al., 2025 JINST 20 P06035

Thanks for your attention!

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